

Topic: Subspaces

Question 1. Suppose $(\mathcal{V}, \oplus, \otimes)$ is a real vector space and $\mathcal{W}$ is a nonempty subset of $\mathcal{V}$. Prove that $\mathcal{W}$ is a subspace of $\mathcal{V}$ if and only if the following two conditions hold:

- (a) $\mathbf{0} \in \mathcal{W}$.
- (b) $(\lambda \otimes \mathbf{u}) \oplus \mathbf{v} \in \mathcal{W}$ whenever $\lambda \in \mathbb{R}$ and $\mathbf{u}, \mathbf{v} \in \mathcal{W}$.

Question 2. Consider the following subsets $\mathcal{W} \subseteq \mathcal{P}_4(\mathbb{R})$ and $\mathcal{V} \subseteq \mathcal{P}_4(\mathbb{R})$, where

$$\mathcal{W} = \{p(x) \in \mathcal{P}_4(\mathbb{R}) \mid p(0) = 0 \text{ and } p(2) = 0\},$$

and

$$\mathcal{V} = \{p(x) \in \mathcal{P}_4(\mathbb{R}) \mid p(0) = 0 \text{ or } p(2) = 0\}.$$

- (a) Is $\mathcal{W}$ is a subspace of $\mathcal{P}_4(\mathbb{R})$?
- (b) Is $\mathcal{V}$ is a subspace of $\mathcal{P}_4(\mathbb{R})$?

Question 3. Consider the real vector space $\mathcal{F}(\mathbb{R}, \mathbb{R})$ of functions on $\mathbb{R}$. Which of the following subset $\mathcal{W}$ is a subspace of $\mathcal{V}$? Justify your answer.

- (a) $\mathcal{W}$ is the subset of all even functions. Recall that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is even if $f(-x) = f(x)$ for all $x \in \mathbb{R}$.
- (b) $\mathcal{W} = \{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) : f(x) = 0 \text{ for all } x \in S\}$, where S is a nonempty subset of $\mathbb{R}$.
- (c) $\mathcal{W} = \{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) : f(x) \geq 0 \text{ for all } x \geq 0\}$.
- (d) $\mathcal{W}$ is the set all differentiable real-valued functions defined on $\mathbb{R}$.

Question 4. Let $\mathcal{V}$ be a vector space. Suppose $\mathcal{W}_1$ and $\mathcal{W}_2$ are subspaces of $\mathcal{V}$. Which of the following subsets are always subspaces of $\mathcal{V}$? Justify your answer.

- (a) $\mathcal{W}_1 \cup \mathcal{W}_2$.
- (b) $\{\mathbf{0}\} \cup \mathcal{W}_1^c$. (Here, $\mathcal{U}^c$ denotes the complement of $\mathcal{U}$ in the set $\mathcal{V}$.)
- (c) $\mathcal{W}_1 \cap \mathcal{W}_2$.

Question 5. Let $\mathcal{V}$ be a vector space. Suppose $\mathcal{W}_1$ and $\mathcal{W}_2$ are subspaces of $\mathcal{V}$. We define the sum of subspaces $\mathcal{W}_1 + \mathcal{W}_2$ as follows

$$\mathcal{W}_1 + \mathcal{W}_2 = \left\{ \mathbf{w}_1 + \mathbf{w}_2 \in \mathcal{V} : \mathbf{w}_1 \in \mathcal{W}_1 \text{ and } \mathbf{w}_2 \in \mathcal{W}_2 \right\}.$$

- (a) Consider the vector space $\mathbb{R}^3$, and its two subspaces $\mathcal{W}_1$ and $\mathcal{W}_2$ which are x -axis and y -axis respectively.

$$\mathcal{W}_1 = \{(x, 0, 0) : x \in \mathbb{R}\} \text{ and } \mathcal{W}_2 = \{(0, y, 0) : y \in \mathbb{R}\}.$$

Find the sum $\mathcal{W}_1 + \mathcal{W}_2$, the intersection $\mathcal{W}_1 \cap \mathcal{W}_2$ and the union $\mathcal{W}_1 \cup \mathcal{W}_2$. (Attempt this part to understand the notation!)

- (b) Show that $\mathcal{W}_1 + \mathcal{W}_2$ is a subspace.
- (c) Show that $\mathcal{W}_1 + \mathcal{W}_2$ contains both $\mathcal{W}_1$ and $\mathcal{W}_2$.
- (d) Show that if $\mathcal{W}$ is a subspace of $\mathcal{V}$ and $\mathcal{W}$ contains both $\mathcal{W}_1$ and $\mathcal{W}_2$, then $\mathcal{W}_1 + \mathcal{W}_2 \subseteq \mathcal{W}$.
(Parts (c) and (d) prove that $\mathcal{W}_1 + \mathcal{W}_2$ is the smallest subspace which contains both $\mathcal{W}_1$ and $\mathcal{W}_2$.)

Topic: Span

Question 6. In the following parts, you are given a real vector space $\mathcal{V}$, a vector $\mathbf{v}$ in it, and a subset S of the vector space. Determine if the given vector $\mathbf{v}$ lies in the span of the given subset S . If it does, find an explicit representation of $\mathbf{v}$ as a linear combination of the vectors in S .

- (a) $\mathcal{V} = \mathbb{R}^3$, $\mathbf{v} = (5, 1, -5)$, and $S = \{(1, -2, -3), (-2, 3, -4)\}$.
- (b) $\mathcal{V} = \mathcal{M}_{2 \times 2}(\mathbb{R})$, $\mathbf{v} = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$, and $S = \left\{ \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \right\}$.
- (c) $\mathcal{V} = \mathcal{P}_3(\mathbb{R})$, $\mathbf{v} = 2 + 3x - 11x^2 - 2x^3$, and $S = \{-1 + 3x - 2x^2 + x^3, -2 + 3x + x^2 + 2x^3\}$.

Question 7. Consider the real vector space $\mathcal{M}_{2 \times 2}(\mathbb{R})$. Let

$$\mathbf{M}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{M}_2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{M}_3 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Show that $\text{span}(\mathbf{M}_1, \mathbf{M}_2, \mathbf{M}_3)$ is the set of all 2×2 symmetric matrices.

Question 8. Suppose S_1 and S_2 are subsets of a vector space $\mathcal{V}$.

- (a) Show that
- $$\text{span}(S_1 \cap S_2) \subseteq \text{span}(S_1) \cap \text{span}(S_2).$$
- (b) Give one example in which $\text{span}(S_1 \cap S_2)$ and $\text{span}(S_1) \cap \text{span}(S_2)$ are equal, and one in which they are unequal.